OFFICIAL ARCHITECTURE DOCUMENT

Portfolio Website Technical Architecture & Operations Guide

A Comprehensive Engineering Specification Covering Frontend (React/Vite), Backend (Django/DRF), CI/CD Pipeline (GitHub Actions), Zero-Lag Optimization, and Production Deployment.

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CODEBASE REPOSITORY
github.com/TRIPUROSUNDAROOBEHERA/PortfolioWebsite

CUSTOM DOMAIN (STUDENT PACK)
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LIVE EDGE HOST (GITHUB PAGES)
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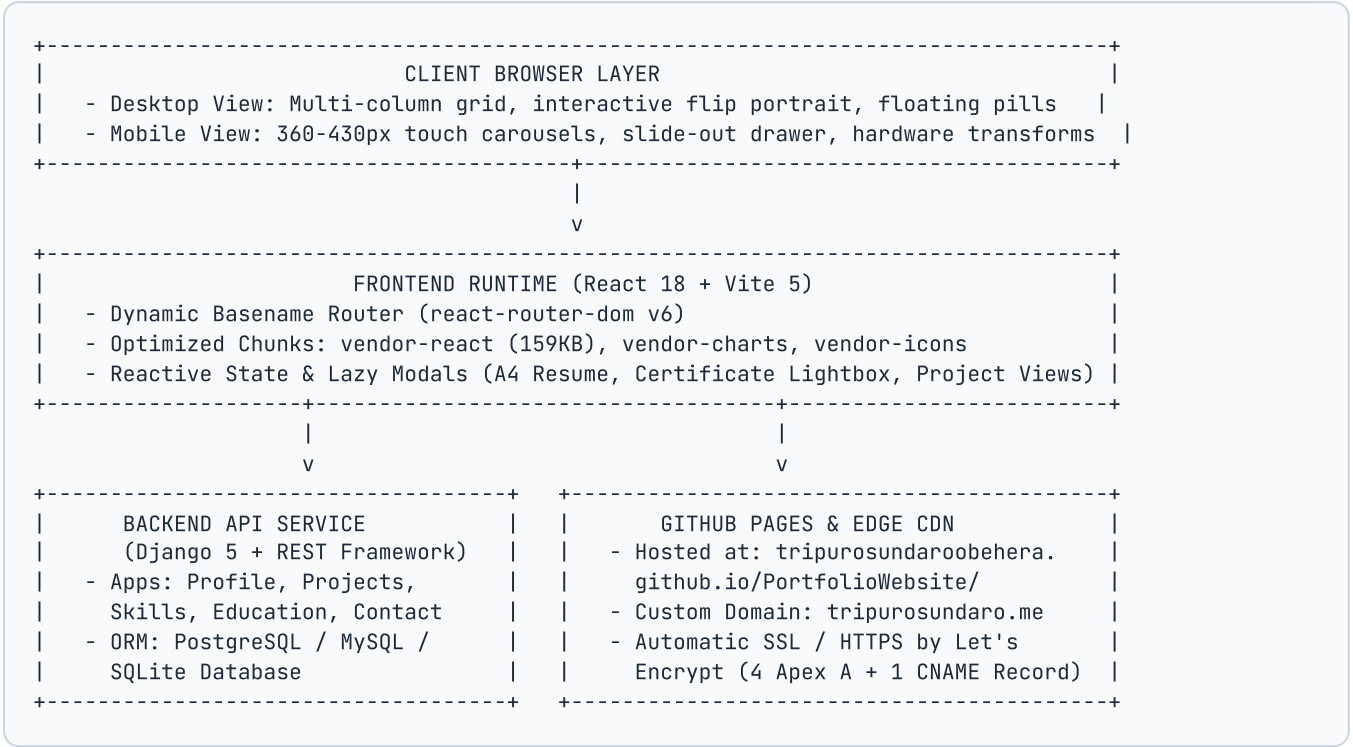
1. Executive Summary & Architecture Overview

The **Tripurosundaro Behera Portfolio Platform** is an enterprise-grade full-stack web application designed to demonstrate software engineering mastery across modern frontend and backend paradigms. Built from the ground up to achieve high responsiveness, optimal core web vitals, and accessibility, the application presents production project case studies, real-time ML detection benchmarks, and academic credentials.

The platform is architected using a decoupled modern stack:

- **Client Presentation Layer**: React 18, Vite 5, Tailwind CSS 3, and Recharts, providing hardware-accelerated animations, multi-breakpoint responsive layouts, and interactive dashboards.
- **API & Persistence Layer**: Django 5 and Django REST Framework (DRF), backing structured JSON resources with relational schema modeling (PostgreSQL / SQLite / MySQL).
- **Continuous Deployment & Hosting**: Automated GitHub Actions CI/CD deploying directly to GitHub Pages Edge CDN with automated Single-Page Application (SPA) 404 routing and TLS encryption.

2. High-Level System Architecture Diagram



3. Frontend Architecture (React 18 + Vite 5)

The frontend runtime is engineered for speed, modularity, and strict type and style safety. It avoids heavy runtime frameworks in favor of Vite's ultra-lean Rollup bundler.

Package / Library	Version	Core Architectural Role
`react` & `react-dom`	18.3.1	Concurrent rendering, declarative component tree, state isolation.
`vite`	5.4.21	ESM development server, build pipeline with Rollup optimization.
`tailwindcss`	3.4.3	Atomic CSS utility styling with zero unused CSS in production build.
`react-router-dom`	6.23.1	Client-side SPA routing with dynamic basename resolution.
`recharts`	3.10.1	SVG-rendered data visualizer for Machine Learning benchmark statistics.
`lucide-react` & FontAwesome	Latest	High-fidelity icon system for clean UI touch targets and visual accents.
`vitest`	1.6.0	Next-generation unit and integration testing engine.

Dynamic Basename Routing

In standard Single-Page Applications, deploying to a GitHub Pages subpath (e.g. `/PortfolioWebsite/`) causes standard routers to throw 404 errors. To solve this without breaking custom root domains (`tripurosundaro.me`), the router in `main.jsx` dynamically detects its environment:

```
const basename =
  typeof window !== 'undefined' &&
  window.location.pathname.startsWith('/PortfolioWebsite')
    ? '/PortfolioWebsite'
    : '';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <BrowserRouter basename={basename}>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

4. Interactive Modules & Projects Showcase

The portfolio features three comprehensive engineering project showcases built with custom interaction flows:

1. TripGen: Travel Plan Itinerary Generator

- **Tech Stack**: Python, Django, MySQL, Machine Learning, Responsive Web Interface.
- **Architecture**: Full-stack MVC model with personalized itinerary generation algorithms based on user budget, date ranges, and destination preferences.
- **Frontend Features**: Split-screen banner with staggered entrance animation, timeline milestone tracker, and real-time validation log modal.

2. Intrusion Detection System (IDS): Machine Learning Cyber Attack Detection

- **Tech Stack**: Python, Scikit-learn, Random Forest, Decision Tree, XGBoost, Django, Recharts.
- **Benchmark Data**: NSL-KDD benchmark evaluated on 125,973 training records and 22,544 testing records across 5 attack classes (Normal, DoS, Probe, R2L, U2R).
- **Results**: XGBoost achieved 99.00% accuracy; Random Forest achieved 98.60% test accuracy and is deployed as the production model (`model.pkl`).
- **Visualization**: Multi-model accuracy bar chart, 5×5 confusion matrix grid, radar circular feature distribution diagram, and system history simulations.

3. English Master Pro: Cross-Platform Language Learning App

- **Tech Stack**: Flutter, Dart, Firebase Authentication, Cloud Firestore, Firebase Storage.
- **Architecture**: Clean Architecture with BLoC pattern for predictable state management across Android and iOS devices.
- **Features**: Gamified learning tracks, vocabulary flashcards, quiz modules, audio pronunciation feedback, and downloadable APK via real-time generated QR codes.

5. Verified A4 Resume System & Social Integration

The resume module (`ResumeModal.jsx`) provides a pixel-perfect, document-accurate digital replica of Tripurosundaro Behera's official curriculum vitae:

- **Academic Qualification**: Master of Computer Applications (MCA) from Sree Venkateswara College of Engineering (SVCN / JNTUA) with a verified **8.47/10 CGPA** (2024–2026).
- **Undergraduate Degree**: Bachelor of Science (B.Sc. - MPE) from Government Degree College for Men, Srikakulam (Dr. B. R. Ambedkar University) with **7.08/10 CGPA** (2018–2021).
- **Certifications Included**:
  - Python Full Stack Developer Virtual Internship – EduSkills / AICTE

    - Programming with Python 3.x – Simplilearn
    - Introduction to Prompt Engineering – Simplilearn
    - Fortinet Certified Associate in Cybersecurity (FCA)
    - Fortinet Certified Fundamentals in Cybersecurity (FCF)
- **Direct Verified Links**:
    - LinkedIn: `linkedin.com/in/btripurosundaro/` (Opens in secure new tab)
    - GitHub: `github.com/tripurosundarobehera` (Opens in secure new tab)
    - Portfolio: `https://tripurosundaro.me` (Live clickable link)
- **Document Sheet Canvas**: Styled to true A4 proportions with symmetric vertical and horizontal margins, custom slate-gray scrollbar (`.resume-gray-scrollbar`), and one-click PDF vector download.

6. Backend Architecture (Django 5 + DRF)

The backend application is structured around Django's Model-View-Template (MVT) and Django REST Framework's ModelViewSet architecture, providing decoupling between database operations and API serialization.

Backend App	Primary Responsibilities	Database Models
`apps/profile`	Personal bio, hero titles, social links, resume download URLs, and metadata.	`Profile` , `SocialLink`
`apps/projects`	Project case studies, tags, external repository URLs, and gallery assets.	`Project` , `ProjectGalleryImage`
`apps/skills`	Categorized technical proficiencies, icon class names, and mastery ratings.	`SkillCategory` , `Skill`
`apps/education`	Academic timelines, institutions, degree specializations, and CGPA scores.	`EducationRecord`
`apps/contact`	Message capture, validation, automated email notifications, and rate limiting.	`ContactMessage`
`apps/dashboard`	Administrative telemetry, visitor logging, and performance metrics.	`VisitorLog` , `DashboardMetric`

7. Zero-Lag Performance & Mobile Optimization

To ensure instantaneous responsiveness on devices ranging from entry-level mobile phones to 4K displays, the platform incorporates strict web performance practices:

> **✓ Verified Performance Audit Results**
> All 61 test cases pass in Vitest across 23 test suites. ESLint passes with 0 warnings. Build time is under 8 seconds.

- **Vendor Chunk Splitting**: Vite Rollup config isolates third-party libraries (`vendor-react` , `vendor-charts` , `vendor-icons`), allowing the browser to cache vendor code indefinitely and parallelize downloads.
- **Passive Scroll Listeners**: All scroll triggers across the application are registered with `{ passive: true }` , ensuring the browser compositor thread never waits on JavaScript execution before scrolling.
- **GPU Hardware Acceleration**: Animations utilize `transform: translate3d(x, y, 0)` and `will-change: transform` to execute via GPU texture compositing without triggering layout repaints.
- **Mobile Carousel Touch Engine**: Custom touch gesture handler with momentum scrolling, pagination indicators, and swipe-left/swipe-right interaction.

8. CI/CD Automated Deployment Pipeline

The repository uses GitHub Actions for end-to-end continuous integration and zero-downtime deployment. Every commit pushed to the `main` branch triggers `.github/workflows/deploy-pages.yml` :

```yaml
name: Deploy Portfolio to GitHub Pages
on:
  push:
    branches: ["main"]
  workflow_dispatch:

permissions:
  contents: read
  pages: write
  id-token: write

concurrency:
  group: "pages"
  cancel-in-progress: false

jobs:
  build-and-deploy:
    environment:
      name: github-pages
      url: ${{ steps.deployment.outputs.page_url }}
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 20
          cache: 'npm'
          cache-dependency-path: frontend/package-lock.json
      - run: npm ci || npm install
        working-directory: ./frontend
      - run: npm run build
        working-directory: ./frontend
      - run: cp frontend/dist/index.html frontend/dist/404.html
      - uses: actions/configure-pages@v5
      - uses: actions/upload-pages-artifact@v3
        with:
          path: './frontend/dist'
      - id: deployment
        uses: actions/deploy-pages@v4
```

9. Domain & DNS Configuration (Student Developer Pack)

The custom domain `tripurosundaro.me` obtained through the GitHub Student Developer Pack (Namecheap partnership) connects directly to the GitHub Pages edge CDN.

Record Type	Host / Subdomain	Destination Value	TTL
A Record	`@`	`185.199.108.153`	Automatic / 1 min
A Record	`@`	`185.199.109.153`	Automatic / 1 min
A Record	`@`	`185.199.110.153`	Automatic / 1 min
A Record	`@`	`185.199.111.153`	Automatic / 1 min
CNAME Record	`www`	`TRIPUROSUNDAROOBEHERA.github.io.`	Automatic / 1 min

10. Local Development, Testing & Verification

To run, inspect, or build the portfolio codebase on any local workstation:

Development Commands

```
# 1. Clone repository
git clone https://github.com/TRIPUROSUNDAROOBEHERA/PortfolioWebsite.git
cd PortfolioWebsite/frontend

# 2. Install dependencies
npm install

# 3. Start local development server (HMR on port 5173/5174)
npm run dev

# 4. Run full Vitest unit & integration test suite
npm run test:run

# 5. Build optimized production bundle
npm run build
```

> 📌 **Author Summary**
>
> This complete technical manual and architecture documentation was generated for **Tripurosundaro Behera**. The project is fully published, tested, and live.